SandEE: AN EDUCATIONAL MULTI-PLATFORM 2D OPEN WORLD SURVIVAL CRAFTING GAME FOR LUA AND PYTHON FOR ICT AND STEM STUDENTS OF YVERDON DE PESTALLOZI SCHOOL INC.

A Thesis Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College San Jose Del Monte

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Computer Science

Glenn Marco B. Accad
Adrian V. Gallano
Jefferson G. Mamauag
Jose Mari Christian V. Tayag

November 2026

ENDORSEMENT FORM FOR PROPOSAL DEFENSE

TITLE OF RESEARCH: **SandEE: AN EDUCATIONAL MULTI-PLATFORM 2D OPEN WORLD SURVIVAL CRAFTING GAME FOR LUA AND PYTHON FOR ICT AND STEM STUDENTS OF YVERDON DE PESTALLOZI SCHOOL INC.**

NAME OF PROPONENTS: Glenn Marco B. Accad
Adrian V. Gallano
Jefferson G. Mamauag
Jose Mari Christian V. Tayag

In Partial Fulfilment of the Requirements
for the degree Bachelor of Science in Computer Science
has been examined and is recommended for Proposal Defense.

ENDORSED BY:

Norvin Shaq F. Lintag
Thesis Adviser

APPROVED FOR PROPOSAL DEFENSE:

Enoch James C. Tayanguna
Thesis Coordinator

NOTED BY:

Shiela Marie B. Llesol
Program Head

November 2026

APPROVAL SHEET

This thesis proposal titled **SandEE: An Educational Multi-Platform 2D Open World Survival Crafting Game for Lua and Python for ICT and STEM Students of Yverdon de Pestallozi School Inc..**, prepared and submitted by **Glenn Marco B. Accad; Adrian V. Gallano; Jefferson G. Mamauag**; and **Jose Mari Christian V. Tayag**, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science, has been examined and is recommended for acceptance and approval.

Norvin Shaq F. Lintag
Thesis Adviser

Accepted and approved by the Thesis Review Panel
in partial fulfillment of the requirements for the degree of
Bachelor of Science in Computer Science

Louis Matthew P. Sindao
Panel Member

Lianoel Vergante
Panel Member

Enoch James C. Tayanguna
Lead Panelist

Noted:

Enoch James C. Tayanguna
Thesis Coordinator

Shiela Marie B. Llesol
Program Head

November 2025

TABLE OF CONTENTS

	Page
Title Page	i
Endorsement Form for Proposal Defense	ii
Approval Sheet	iii
Table of Contents	**iv**
Introduction	**6**
Background of the problem	8
General Problem	9
Specific Problem	9
Overview of the current state of the technology	11
Objectives of the study	14
General Objectives	14
Specific Objectives	14
Scope and limitations of the study	16
Review of related literature/studies/systems	30
Foreign Literature	30
Local Literature	31
Foreign Studies	32
Local Studies	34
Foreign Systems	35
Local Systems	36
Synthesis	37
METHODOLOGY	39
Hardware/Software	41
Calendar of Activities	43
Budgetary Estimate	45
Human Resources	46
Survey	51
ADVISER'S ACCEPTANCE FORM	**58**
References	**59**

INTRODUCTION

In recent times, especially in information communication technology, programming has been the focus of computing education. The recent advent of several tools has made coding more accessible than ever (Becker, 2023). Programming has been a challenge for learners. The proliferation of digital technologies has made programming skills increasingly important across various disciplines. This shift extends beyond traditional computer science domains (Lin, et.al., 2025). According to industry reports, the demand for Lua developers has been steadily increasing in recent years. As more companies turn to Lua for its speed, simplicity, and versatility, the need for skilled Lua developers has never been higher. In particular, the gaming industry has seen a rise in Lua-related job openings, with many companies looking for developers proficient in Lua scripting (Crudu, 2024). The Lua scripting language is a fast, procedural programming language it was designed to be highly customizable with simple API in the C programming language so that programmers could easily make changes to fit their needs. Python has a simple syntax that allows developers to write programs with fewer lines compared to other programming languages Additionally, Python as a scripting language has been steadily gaining traction since its inception. Its simplified syntax allows learners to easily learn and grasp the coding structure of the scripting language (Python, 2021).

Yverdon De Pestalozzi, Inc. is a private senior high school offering STEM and ICT strands, dedicated to developing students' technical skills, academic competence, and readiness for future careers through a supportive and well-structured learning environment.

According to Lampropoulos (2023), Game-based learning emerged as a useful education approach that can be facilitated and enrich the learning process especially in students. This approach creates a motivational virtual learning environment that consciously engages and encourages students to actively participate in educational activities. This approach has also been used extensively in the field of programming, allowing an alternative approach to learning various programming concepts.

In a survey conducted to ICT and STEM Senior High School students of Yverdon De Pestallozi School Inc., a total of 104 respondents participated, consisting of 22 students from Grade 11 ICT (21.2%), 32 from Grade 11 STEM (30.8%), 16 from Grade 12 ICT (15.4%), and 34 from Grade 12 STEM (32.7%). The results revealed that most students demonstrated avergae familiarity with the Lua programming language, suggesting that it is relatively new to many respondents. In addition to this, the majority expressed positive attitudes toward learning Lua through supplementary game-based activities, with 81 (78.9%) agreeing that such an approach would improve their understanding of programming concepts and 76 (73%) showing motivation to explore Lua using an interactive approach. On the other hand, Python received generally favorable feedback, with 84 (80.8%) of participants finding its syntax and structure easy to understand and applicable to programming tasks. Overall, the findings indicate that while students are less experienced with Lua compared to Python, they show strong enthusiasm and willingness to learn it through interactive and game-oriented methods that enhance engagement and learning outcomes. As a result, introducing a gamified approach of learning Lua and Python is applicable to the students at the school.

SandEE is an educational 2D open world survival crafting game for Lua and Python. It introduces several programming concepts related to Lua and Python. Furthermore, SandEE also includes gameplay that can serve as an exercise to advanced users of Lua and Python. The proposed study offers a supplementary learning approach for Lua and Python by slowly introducing its syntax and structure. The game embeds interpreters for Lua and Python in which code that is run impacts the game environment to show visual feedback, as well as to provide a realistic basis to the player as the interpreters will give information accurate to its real use-case. Progression is achieved through the mission system, which guides the player to progress in learning. These missions will eventually lead to rewards to further gratify the player and give them feedback on their progress.

Background of the problem

The rapid advancement of technology has significantly transformed educational methodologies. Traditional learning, which often emphasizes theoretical concepts and static exercises, struggles to fully engage students or prepare them for real-world problem-solving. In contrast, educational games have emerged as an innovative approach to learning, combining interactivity and engagement with skill development. Popular examples like Duolingo demonstrate how gamified systems can make learning both effective and enjoyable. Game-based learning in programming has shown the potential to enhance logical reasoning, problem-solving, and creativity through active participation.

To better understand the learning preferences of ICT and STEMS students, a survey was conducted among the grade 11 and 12 ICT and STEM students of Yverdon De Pestallozi School Inc. The results revealed that most respondents had average familiarity with the Lua programming language, with the majority rating their understanding between 2 and 4 on a 4-point scale. However, despite this lack of experience, 82 (78.9%) of respondents rated 2, 3, and 4 when asked how helpful and motivating it would be to learn Lua through game programming. For comparison, students showed higher familiarity with Python, averaging between 2, 3 and 4 in both ease of use and coding level. Yet even those comfortable with Python expressed strong interest in learning programming concepts through interactive, game-based methods. These findings indicate that students are more receptive and motivated when learning involves playing, which shows what happens inside the game, experimentation, and engagement rather than passive listening and reading books.

To address these findings, the proposed game introduces an immersive, game-based platform that integrates survival and crafting mechanics within an open-world sandbox-type environment. Through hands-on coding challenges embedded in gameplay, it aims to foster critical thinking, logical reasoning, and practical programming skills. This approach seeks to make programming educational, creative, experiential, contextual, and enjoyable.

General Problem

How to develop a game to supplement the high interest in learning Lua and Python through a game?

Despite low familiarity with ICT and STEM students, survey results have shown that interest in learning Lua and Python is high. Furthermore, the results have also shown an increasing interest in learning these languages using a more interactive and engaging approach through a game.

Specific Problem

- **How to develop a game that enhances programming skills by providing hands-on learning where concepts are learned through experience?**

  Many ICT and STEM students have shown a strong interest in learning programming languages such as Lua and Python, particularly through interactive and engaging methods. Based on the survey conducted, there is a clear inclination toward learning through game-based experiences that allow the practical use of programming concepts. Therefore, this study focuses on developing a game that enhances programming skills by providing hands-on learning where concepts are learned through experience, serving as an innovative approach to support growing interest in programming.

- **How to develop a game that provides direct visual feedback as players learn and apply programming concepts?**

  Students often show greater engagement and understanding when they can immediately see the results of their actions while learning. In programming education, providing direct visual feedback allows learners to observe how their inputs affect outcomes in real time, creating a more interactive and immersive experience. Therefore, this study aims to develop a game that provides direct visual feedback as players learn and apply programming concepts, promoting better comprehension and active participation through visual interaction.

- **How to develop a game that realistically responds to player errors by embedding Lua and Python interpreters within the system?**

  To create a more authentic and educational programming experience, it is important for a game to respond realistically to player inputs, including coding errors. Embedding Lua and Python interpreters within the system enables real-time evaluation of player code, allowing the game to simulate how actual programming environments handle mistakes.

- **How to develop a game that introduces the programming structure and syntax of Lua and Python?**

  Learning programming languages such as Lua and Python requires a clear understanding of their syntax and structural rules. Introducing these foundational concepts through an interactive approach can make learning more engaging and effective.

Overview of the current state of the technology

In recent years, advancements in teaching coding through educational and engaging means have been accomplished. There are ways, including game-based learning and digital courses, that provide plenty of valuable knowledge on how to code to solve problems and perform tasks.

Philippine Coding Camp (2023)

Such examples include the Philippine Coding Camp. These are digital learning bootcamps that primarily focus on teaching students of any age by making the courses simpler, welcoming, and open to beginners and experienced learners alike. Both provide structured courses that teach the fundamentals and concepts of programming, which they can later apply to what they have learned through exercises and challenges. They use platforms such as Roblox as a supplement and make learning more engaging. As learners progress through the courses, new concepts are slowly and simply introduced for them to better understand and retain what they are learning.

While these courses are structured, there are limitations that can be addressed. Such as the focus is only centered on the topics of the course. By making the process more hands-on, it removes the restriction of complying with the structure of the course and letting the learners learn in their own way with similar guidance.

Logiscool (2025)

Logiscool is a similar coding camp established worldwide and has a digital platform where they also teach digital courses. They provide programs where in each, they teach digital literacy, AI, and coding. They integrate gaming platforms such as Roblox and Minecraft into their learning curriculums to make learning more interesting. In addition, they also use artificial intelligence to support teaching. Part of the programs they teach are emerging languages such as Python and Lua. Python is taught within topics such machine learning, and Lua is primarily used in Roblox to create games and experiences. Logiscool's main target is younger users but are still open to everyone.

Logiscool's programs thoroughly teach its learners about the topics, provide and hones skills such as critical thinking and creativity. But by creating a more engaging way to teach, similarly to how they use games, learners can better learn and apply their skills into practical applications.

Hello World (2024)

Hello World is a game that primarily teaches Java and C#. It teaches fundamental concepts of these two-programming languages and integrates them into gameplay. Throughout Hello World, introductory lessons are provided, then the player must use what they have learned to progress. The game then adds the challenge of using different concepts to solve a given problem. It teaches through direct engagement with languages since it is incorporated into the gameplay. The strength of the system comes from the interactive process of learning languages.

This study provides aims to recreate the process of learning through interaction with languages, but uses two emerging languages, Lua and Python. Java and C# are already commonly taught in the field, but there is a growing need in the industry that seeks developers with knowledge of Lua and Python.

Code Combat (2025)

CodeCombat is a digital gaming platform that primarily focuses on teaching fundamentals of programming and applying it into game scenarios. Ozaria is built from CodeCombat, where Python is one of the main focuses. They have a custom built-in engine and interpreter that helps beginners in learning how to code. This platform can be integrated in school settings, allowing students to share their projects with others and teachers to grade their work. CodeCombat works by teaching students first, then placing them in game scenarios where their knowledge is put to the test. Ozaria expands this by creating a storyline that progresses with the learner's skill and knowledge on the subject.

There are limitations to this, however. While this type of learning is more engaging, it lacks the structured format that courses provide. They teach first then send the learners into performing tasks. The study aims to bridge the gap between digital courses and gamified learning by blending the two ways of learning into a consolidated process. By merging the structured format of courses and the hands-on engagement that games provide, this creates a foundation for students to learn programming concepts in a unified form.

CodinGame (2025)

CodinGame is an online gaming platform that uses coding as its main method of playing. It challenges developers to use their skills in competing against others across different types of games. It encourages the application of what the user knows about programming. It features several programming languages such as Lua and Python, and developers must utilize their skills of the language throughout CodinGame's challenges and competitions. This type of learning helps developers see what they can still learn and improve to solve future challenges and tasks they may face.

CodinGame is limited since it primarily focuses on competition against other players. It doesn't provide an exact way to teach; it only points out what is lacking when it comes to knowledge. This type of learning is limited, hence creating a game that

teaches, and rewards players gives a sense of accomplishment and progress that provides a better experience compared to the competitive nature of CodinGame.

Objectives of the study

General Objectives

To develop a game to supplement the high interest in learning Lua and Python through a game.

The proponents of the projects aim to develop a game with the goal of supplementing the high interest of learning the Lua and Python programming language through a game approach by integrating engaging gameplay with learning activities.

Specific Objectives

- **To develop a game that enhances programming skills by providing hands-on learning where concepts are learned through experience.**

  The proponents of the project aim to create an interactive learning environment where students can apply programming concepts directly within gameplay. Specifically, the game will encourage learners to solve coding challenges that simulate real-world scenarios, promote active problem-solving, and experiential learning rather than passive observation.

- **To develop a game that provides direct visual feedback as players learn and apply programming concepts.**

  The proponents of the project plan to use the game's environment to display visual feedback based on the user's coding input. Specifically, the player's actions and written code will directly affect the environment, allowing them to see how Lua and Python interpret their logic in real time. This feedback helps learners understand programming flow and identify errors through immediate, intuitive results.

- **To develop a game that realistically responds to player errors by embedding Lua and Python interpreters within the system.**

  The proponents of the project aim to ensure that the game reacts authentically to user errors by integrating Lua and Python interpreters. Specifically, this enables the application to detect and display syntax or runtime errors accurately, providing a realistic debugging experience. This approach helps students learn from their mistakes and develop stronger problem-solving and error-handling skills.

- **To develop a game that introduces the programming structure and syntax of Lua and Python.**

  The proponents of the project intend to integrate Lua and Python learning modules within the game. Specifically, the application will include coding activities that demonstrate syntax rules, control structures, and logical operations in both languages. This allows users to understand and practice programming concepts naturally while progressing through the game.

Scope and limitations of the study

- **User Account**
    - Each user can create a unique account using their own username and password.
- **Cloud Saves**
    - The save file of each user will be uploaded to the internet every time the game saves.
    - The save file of each user is accessible multi-platform.
    - Cloud saves prioritize the save with the highest amount of stars to prevent conflict.
- **Offline Mode**
    - Allow the player to play the game offline.
    - Players can switch to online mode from offline mode.
- **Online Mode**
    - Allows the player to access the leaderboard and cloud saves.
- **Main Menu**
    - Continue
        - This allows the player to continue from their recently saved world.
    - Worlds
        - This will redirect the user to the saves menu to select their desired save or start a new save.

- Leaderboards (Online Mode Only)
    - List the rankings of player's acquired blue and yellow stars.
- Cloud Save Management (Online Mode Only)
    - Allow the player to manage cloud saves.
- Account (Online Mode Only)
    - Allow overview of the account and allow logging out.
- Settings
    - This will redirect the user to the settings panel to adjust the game's sound volume.
- Quit
    - This will close the game.

- **Settings Panel**
    - Volume Sliders
        - BGM Volume
            - Adjusts the volume for the background music of the game.
        - Effects Volume
            - Adjust the volume for the sound effects of the game.
    - Fullscreen
        - Sets the game to full screen window.
    - Credits

        - Lists the developers and individuals who are part of the game development along with references to the materials and software used.

- **Worlds Menu**
    - World Creation
        - Lets the player name the world before creation.
    - Delete
        - Deletes the chosen save file upon confirmation.
    - Rename
        - Allows the player to rename their worlds.
    - Save Slot
        - Loads the chosen save file to continue its progress.
- **Game Mechanics**
    - Procedural Map Generation
        - Single player maps are automatically created using algorithms instead of being manually designed which allows each environment to be unique.

- Missions
    - A list of gameplay objectives that gradually guides the player throughout the entire game.
    - Dungeon missions are separated into distinct levels of difficulties which encompasses a gradual flow of progression into learning programming.
        - Easy mission introduces basic concepts.
        - Medium missions combine concepts.
        - Hard missions teaches advanced concepts.
    - Completing missions will give the player stars and the stars they have will determine their rank.
- Ranks
    - Ranking system will allow players to display their programming capabilities in the game.
    - Ranks can be increased by completing dungeons and earning blue stars.
- Completion Badge
    - Granted to the player upon completing extra dungeon missions. Only given once.
- Leaderboards
    - Sort available players by their rank.
    - Rankings are based on the blue stars the player has.

- Crafting
    - The user will be crafting various items that improve their current state or progress the missions provided.
    - The user can craft multiple categories of items which include materials, interactive objects, weapons, and foods. Some of which can have additional functionality using programming.
    - The users can craft items to build and progress.
- Fishing
    - Players can use fishing rods to catch fishes in bodies of water.
- Farming
    - Players can use tools to collect seeds and build farmland near water.
- Items
    - Materials
        - It can be used to craft materials and other craftable items.
    - Interactive Objects
        - Crafting Stations
            - Where the user can craft items.
        - Furnace
            - Allows the player to smelt ores and cook food using either wood or charcoal as fuel.

- Chest
    - Allows the player to store items in a limited capacity.
- Color Crafter
    - Allows the player to craft items with custom RGB colors.
    - Displays the hex number for the selected block.
    - Player can select between hex or RGB.

- Weapons
    - Swords can be used to deal damage to enemies.
- Tools
    - Tools such as the pickaxe, axe, and fishing rod can be used for specific purposes.
- Food
    - Can be used to heal health.

- Dungeons
    - Randomly generated areas that serve to give major progress for the user, challenging their skills in Lua or Python programming to unlock more mechanics.
    - Rewards the player with blue stars upon completion.
    - Serves as a guide to player's progress.

        - Randomly spawns successively throughout the world and serve as the dungeon mechanic of the game. The next dungeon spawns after the last one are completed.
        - Includes quizzes that can be solved openly depending on the user.
        - Features Mini-Boss Battle in the last stage.
    - Caves
        - Large Caves
            - Generates in the overworld. Provides finite source of stone and ores.
        - Pixie Caves
            - Special caves with resources that regenerate after a period of time.
        - Friendly Mobs
            - Passive mobs like sheeps and wolves wander the overworld.
    - Enemies
        - Spawns randomly on the overworld and in dungeons.
        - Enemies that spawn on dungeons do not adhere to the day and night cycle.
        - Headless Fish
            - Spawns in the overworld during the night and attacks by homing and dashing into the player.

- Desert Worm
    - Spawns in the overworld regardless of daylight. Spawns and hides in the desert and attack the player when approached.
- Hitman
    - Spawns in the dungeon. Fires projectiles and can charge straight into the player to deal damage.
- Pixie
    - Spawns in the Pixie Caves and attacks the player on sight.

- Day and Night Cycle
    - Determines enemy spawning rate and difficulty.
    - Enemies deal less damage during daytime and do normal damage at night.
- Live Interpreter
    - Serves as a built in IDE for the coding function in the game for both Algorithmica and Automatica.
    - Syntax Highlighting which will highlight the errors in programming syntax.
    - Interprets the code that is written by the player.
- Automatica
    - Robot-like craftables that allow the user to automate certain processes by programming it through Lua, through the Lua-focused dungeons.
    - Automaticas can be used to automate interaction with objects and aid the player in combat.
    - Automaticas required placed wires to interact with stations.
    - Automaticas have the option to be renamed.
- Algorithmica
    - Allows the user to automate actions in crafting stations, furnaces, and color crafters.

- LUA (Live Unit Application)
    - An advanced Automatica non-player character that introduces the user to the Lua language.
- Py-san
    - An advanced Algorithmica non-player character that introduces the user to the Python language.
- Autosave
    - The game will automatically save the player's progress periodically and during state changes such as entering caves and dungeons, as well as when the player quits the game.
- Build Mode
    - Can be toggled on or off to place down blocks.

- **Desktop and Mobile Application**
    - HUD (Heads-Up Display)
        - Pause
            - Allow the user to temporarily pause the game, showing the Pause Menu Panel.
        - Pause Menu Panel
            - Resume
                - Resumes the current game.
            - Settings
                - Shows the settings panel.
            - Lua Management
                - Allows the player to manage the LUA robots.
            - Quit
                - Saves the game and directs the user to the Start Page.
        - Mission Panel
            - Shows the user the current missions they can do to progress.
        - Health Bar
            - Shows the user their current health.
        - Stamina Bar
            - Shows the current stamina, which drains when running.

- Hotbar
    - Show the user their current held items.
- Inventory
    - Shows the user their extra space and a smaller crafting area.
- Programming Encyclopedia
    - Contain details about programming concepts for Lua and Python learned through dungeons and custom functions for Algorithmica and Automatica.
- Code Editor Panel
    - A panel that prompts the user to code in Lua or Python to either progress in a dungeon or customize their Automatica and Algorithmica.
- Crafting Panel
    - A panel for more advanced crafting, unlockable when the user crafts the crafting station.
- Map
    - A representation of the world that shows the overworld, cave, and dungeons.

- On-screen Joystick
    - Allows movement for mobile users.
- Interact Button
    - A general button that changes depending on the items near the player's character.
- Build Mode Button
    - Can be toggled on or off if the player wants to build.

Limitations

- **The game only runs on Windows 10 – 11 and Android 8.0 and above.**

  Ensures the game performs consistently across devices with modern hardware and software support. Older systems may lack the necessary performance, security updates, or engine compatibility required by Godot.

- **The game only supports Lua and Python for its languages.**

  This limitation is based on both practicality and accessibility. Lua and Python are in high demand across education, game development, and general programming, making them valuable languages for learners to practice. By restricting the system to these two languages, the game maintains a focused learning environment that is accessible to new programmers while still relevant to real-world applications.

- **The game only supports single player.**

  This limitation ensures that each student's learning experience remains individualized and self-paced. By focusing on single-player gameplay the game allows students to develop their skills independently and maintain consistent progress without distractions from multiplayer interactions.

- **The game only references the 2025 version of official documentation for Lua 5.1 and Python 3.11.**

  This limitation ensures that all learning materials are consistent, stable, and aligned with a single verified source. Lua 5.1 and Python 3.11 have well-established, mature, and consistent syntax, making them reliable for teaching without frequent changes or breaking updates. By locking the game's references to the 2025 documentation versions, students receive clear and uniform guidance.

Review of related literature/studies/systems

Foreign Literature

According to Crudu (2024), Lua developers are in high demand because it is currently widely used in the programming industry. Lua is efficient, flexible, scalable, and cost effective to apply in any programming field. According to the TIOBE Index, Lua is currently ranked as the 32nd most popular programming language. Furthermore, in a survey conducted by Stack Overflow in 2024 it is among the top 10 most loved programming languages.

According to Becker (2022), Generating high-quality learning resources is very time consuming and typically requires a high level of expertise therefore, games have been used as a learning tool for centuries; they have become more prominent in educational settings. When it comes to learning, being focused and immersed can massively improve a student's experience.

According to Viduka (2021), Python is preferred by students and professionals alike due to its ease of use, availability, efficiency, portability, and extensibility. It is a common choice suitable for education. Many schools starting from primary to secondary, all the way to universities prefers teaching Python as primary programming languages.

Local Literature

Computer science software development evolves rapidly; innovative learning platforms transform how instructors teach and how aspiring developers learn programming. Several other solutions integrate interactive web based on learning programming languages. However, these methods of studying are not applicable to everyone. This is where a game-based solution can complement the traditional way of learning (Horlador, et., 2024)

The rapid development of computer science education signifies that traditional methods of learning are being challenged. Conventional approaches often are lacking when it comes to engaging students or providing practical application of programming concepts (Horlador, et., 2024). Traditionally, a beginner developer sits inside a classroom, listening to an instructor as they explain related programming concepts. However, there are instances wherein the beginner developer cannot follow along with the instructor's sample codes (Horlador, et., 2024).

The current education system is outdated in the sense that it cannot keep up with the demands of a fast-paced and information-rich world outside of the classroom. In response to these challenges, innovative learning platforms can be applied to transform the way beginner developers usually learn (Horlador, et., 2024).

In today's rapidly evolving landscape, mastering the concepts of Information Technology is crucial for the students aspiring to enter the technology industry (Abat, et al., 2024). Software development is fundamental to the innovation and efficiency of the applications and solutions tailored to specific needs and demands (Abat, et al., 2024). This caused the current curriculum to include programming languages such as Lua and Python (Abat, et al., 2024).Despite the abundance of academic materials and tutorials regarding different types of programming languages, most of the students still face difficulties in learning the fundamentals of programming and how to apply them. By implementing interactive game-based learning, it allows students to learn using an alternative method of learning that is both fun and educational. (Abat et al., 2024).

Foreign Studies

According to Eltomi (2025), the use of Lua programming language allows the game developers to easily implement core gameplay mechanics such as character movements, physics, and collision detection. This study also identified the problems that game developers encountered such as no prior knowledge in Lua programming language, Game size and complexity, and bad game designs. In relation to the study, it explains the use of Lua programming language in game development. However, the proposed study lacks the idea of showing how the actual Lua language is used in game development.

In a study done by Verbovetsksvi (2025), The authors identified the basic requirements for a game application, such as relevance to educational objectives, incorporation of game mechanics, the ability to write and debug code in the application, individualized passing of the game, and the ability to report and rate learning outcomes of students. The paper contains some fragments of the game workspace and code of a gaming application.

Several studies compare the use of Python and Lua in game development, highlighting their distinct advantages. Python, with libraries like Pygame, is favored for its simplicity and extensive community support, making it ideal for rapid prototyping and educational game development (Oleksiuk et al., 2024). However, it tends to have slower performance compared to Lua. Lua is widely used in professional game engines due to its lightweight nature, fast execution, and seamless integration with C/C++ code, making it suitable for performance-critical game scripting (MoldStud, 2025).

Tyagi's study (2021) compares developing a 2D Flappy Bird game in both Python (Pygame) and Lua (LOVE engine), showing that while Python is easier for beginners, Lua provides better performance and embedding capabilities for commercial projects. Other research agrees that Lua excels in embedded scripting and resource-sensitive environments, whereas Python is often used for game prototyping, tooling, and educational purposes.

In conclusion, these studies demonstrate that the choice between Python and Lua depends on the project goals, with Lua being preferred for efficiency and embedding, and Python for ease of use and broad application.

Local Studies

According to a study done in the University of Mindanao-PALMA Cluster Campus Philippines aimed to determine the effectiveness of game-based learning, based on the findings it can be affirmed that implementing game-based learning strategies is highly effective, resulting in a notable enhancement of learners' academic performance compared to conventional teaching methods. Moreover, the results encourage teachers to revisit their instructional materials and integrate game-based learning materials aligned with specific learning objectives. Additionally, the study provides substantial evidence that game-based learning significantly enhances academic performance compared to traditional instructional resources (Cezar, 2024).

According to Cezar (2022), a study was done to measure the effect of digital game-based learning. Educators and authors argued that video games are constructed on well-developed theories to engage players. The gathered data were interpreted and analyzed; the results revealed that the experimental group achieved higher scores in their academic performance test. The findings showed a significant difference in the performance for students engaging in digital game-based learning.

According to Justo et al. (2022), a study was done to analyze students' perspectives of game-based learning. The results have been proven the game-based learning has a more significant influence on retention and class interaction when this technique is used. As a result, the findings say that game-based learning aids information acquisition and boosts student engagement through increased class interaction.

Foreign Systems

CodeCombat (2024) is a game-based learning platform that teaches languages such as Python through interactive, role-playing gameplay. Its custom-built coding engine and interpreter make it beginner friendly. The platform includes project-sharing features and teacher grading systems, making it usable in both personal and educational contexts. According to Ahlstrom (2025), the system not only simplifies coding, but they execute it in a method that emphasizes having fun. CodeCombat demonstrates how games can effectively serve as a medium for learning programming concepts.

OmegaCrafter (2025) is an open-world survival craft game set in an underdevelopment game world that is being attacked by a mysterious program. Grammi, the players' programmable sidekick can be programmed with various actions and tasks to help with mining, logging, moving materials, and crafting items (Preferred Networks, 2025)

The Farmer Was Replaced (2023) is a game for Python-like language to fully automate various farming tasks that would otherwise be time consuming. (Herzog, 2023). Unlike most programming games, it is not divided into distinct levels that players have to complete but instead features continuous progression. Programming is done in a simple language similar to Python. The game is designed to provide basic programming concepts to the player (Herzog, 2023).

Local Systems

Philippine Coding Camp (2023) is a digital learning center that teaches Filipino students how to code through structured, hands-on programs. The system provides learning courses which teach Lua as one of the foundations for beginner and younger students while also introducing Python later as a part of the game. In relation to the study, the system highlights accessibility in programming education while sustaining engagement and interactivity that traditional digital courses may not fully achieve.

Logiscool (2025) is a digital learning platform that focuses on coding and digital literacy education. Its courses integrate popular platforms to provide engaging environments to enhance coding concepts. The system demonstrates the effectiveness of combining fun and interactivity with education.

Hello World (2024) is a game that teaches the fundamentals of Java and C#. Players apply their skills into gameplay where challenges are introduced. The game progresses based on what the players have learned so far, from introduction to more advanced topics. It is a practical application of game-based learning in the growing industry of programming.

Synthesis

The reviewed foreign and local literature, studies, and systems collectively emphasize the growing importance of game-based learning in programming education. Foreign authors (Becker, 2022; Crudu, 2024; Viduka, 2021) highlighted how games foster immersion and focus, while also pointing to the industry relevance of programming languages such as Lua and Python. Local studies (Horlador et al., 2024; Abat et al., 2024) reinforced these ideas by showing how interactive, game-based methods can complement or even surpass traditional classroom instruction, especially in engaging students who struggle with conventional approaches. Moreover, both foreign and local systems (e.g., CodeCombat, 2024; Philippine Coding Camp, 2023) demonstrated practical applications of integrating coding into interactive platforms, providing evidence that game-based solutions are not only feasible but effective in real-world educational settings.

However, despite these advances, the reviewed literature also reveals a gap: while many systems teach Python or Lua through games, few explicitly focus on bridging the fundamentals of programming with practical coding immersion tailored for beginners in the Philippine context. This gap underscores the significance of the present study, which seeks to design a game-based solution that leverages the strengths of Lua and Python while addressing the challenges faced by novice developers in traditional learning environments. In the context of learning, both data gathered from foreign and local literature are that gaming tools have been used as a powerful supplementary tool for learning and can be applicable into learning complex topics such as concepts of programming (Becker, 2022; Horlador, et al., 2024).

The difference between the gathered information from foreign and local literatures are that in foreign literature, it has been found that games have been increasingly prominent in educational settings since when it comes to learning, students can be more focused and immersed (Becker, 2022). Local literature on the other hand has found that traditional methods are becoming outdated and are challenged by more innovative means of learning, especially considering the context of teaching complex topics in

Computer Science (Horlador, et al., 2024).

Foreign studies done by authors (Eltomi, 2025; MoldStud, 2025; Tiyagi, 2021) gathered that Lua and Python are applicable in implementing features much easier due to each of the advantages that the two languages have. Lua is a lightweight language and integrates with C and C++ code while Python is more beginner-friendly and applicable for prototyping and tooling.

Local studies proven by authors (Cezar, 2022; Justo, et al., 2022) found that gamified learning is an effective method of learning as it has been gathered that the group that was engaged in gamified learning scored better in comparison to the group that did not. Based on what the study has gathered from the local and foreign studies, the goal is to implement gamified learning into teaching students the concepts of Lua and Python as these two languages have been shown to be appealing for beginners to learn (Oleksiuk, et al., 2024; MoldStud, 2025).

The proposed game aims to be unique by implementing the methods of gamified learning into teaching the concepts of Lua and Python in such a way that it acts as a supplement alongside traditional learning methods. It acts to teach students in a more engaging way. It does so by applying concepts and ideas that the foreign and local literature and studies have gathered and combine them in such a way that it introduces a specialized method of teaching students various programming concepts. Lua and Python were chosen as the two programming languages to teach because of their simplicity and utility, making them appealing to teach.

METHODOLOGY

Waterfall Development Life Cycle

Waterfall methodology is a well-established project management workflow. Each process phase cascades downward sequentially through five stages, requirements, design, implementation, verification, and maintenance.

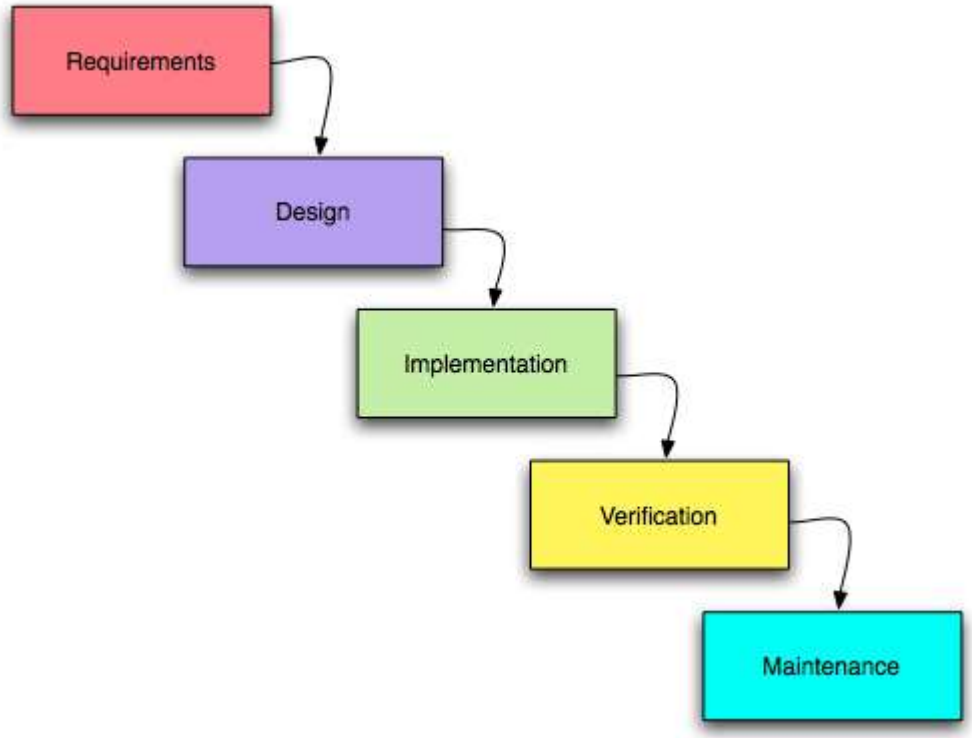


Figure I. Waterfall Methodology

Requirements

The requirement phase states what the system should do. At this stage the project of the scope is determined alongside the resources for the project, allocation of work for each member, timeline of the project, and the details of each stage.

Design

After gathering all the requirements, the developer develops solutions that meet the requirements. This phase focuses on the overall blueprint of the software including its user interface and architecture.

Implementation

Once the design is finalized and approved, the developers build from the specifications of design and collect research needed for the development.

Verification

The verification stage is dedicated to quality assurance and testing of the game alongside fixing the possible errors and bugs that may appear during development.

Maintenance

After the initial release, the developers address the additional bugs that may appear and continue to provide fixes that have not appeared during the initial verification stage. This includes fixing incompatibility issues provided within the limitations of the software and fixing bugs that occurs inside the game.

Hardware/Software

Table I. Hardware

Devices	Specifications
Desktop	CPU: AMD Ryzen 5 5600G Motherboard: MSI PRO B550M-P GEN3 GPU: Radeon Vega 7 Integrated Graphics RAM: 16 GB Operating System: CachyOS (Arch Linux)
Laptop	CPU: i5-12500h Motherboard:AN515-58-50YE GPU: RTX 3050M Ram: 8x2 DDR4 2400mhz sodimm Operating System: Windows 10 Pro

Table II. Software

Software	Purpose
Godot Engine	Game Engine
GDScript	Game Logic
Supabase	Game Database
Aseprite	Art Editing Software
Rust	Performance critical implementation
Lua	Main Scripting Language
Python	Sub Scripting Language
VS Codium	Chosen IDE.
GitHub	Version Control Software, Logger, File Storage

Calendar of Activities

During the first week of September, the developers began working on the initial draft of the study's introduction and finalized the research title. This phase focuses on identifying the study's purpose, significance, and scope while aligning the objectives with the project's intended outcome. The week concluded with the team reviewing the overall direction of the research to ensure that it addressed relevant academic and industry needs.

In the second week of September, the team revised the introduction to improve clarity and alignment with the research objectives. They also began drafting Chapter I, which detailed the background of the problem, statement of the problem, and significance of the study. The group collaborated closely to ensure that the content presented a strong foundation for the succeeding chapters.

The third week of September was dedicated to additional revisions and refinements of Chapter I. The developers reviewed the document to ensure consistency, coherence, and adherence to academic standards. Feedback from peers and advisers was considered to enhance the quality and flow of the content, preparing the chapter for final review.

During the fourth week of September, the developers conducted the survey to gather data from the target respondents. The collected information served as the basis for understanding user preferences and needs. In the same week, the team began drafting Chapter II, which included the review of related literature, studies, and systems relevant to the proposed project.

The first week of October was devoted to extensive research on literature, existing studies, and software systems related to game-based learning. The developers consolidated and analyzed this information to strengthen Chapter II. By the end of the week, Chapter II was completed and prepared for initial revision.

In the second week of October, the proponents focused on drafting the hardware and software selection section. This included identifying the tools, technologies, and development environments necessary for system implementation. The week also involved outlining the technical specifications and requirements for the proposed application.

During the third week of October, the team concentrated on revising and improving Chapter III, which presented the project's methodology. The developers refined the research design, data collection methods, and system development framework to ensure accuracy and alignment with research objectives. In the fourth week of October, the proponents carried out final revisions and grammar checks across all chapters to ensure consistency and correctness. The team also completed the initial design of the software user interface, marking the transition from the documentation phase to the beginning of system development.

Budgetary Estimate

Table III. Budgetary Estimate

Quantity	Specifics	Approximate Cost
1	Acer Nitro 5 AN515-50YE	PHP 49, 000
1 1 1 1 1	Ryzen 5 5600g Asus Prime B550 16 GB ram 1TB SSD 650w PSU	PHP 22, 295

Human Resources

Name: Norvin Shaq F. Lintag

Position: Thesis Adviser

Name: Enoch James C. Tayanguna

Position: Thesis Coordinator

Name: Deffrey Josh A. Austria

Position: Program Head

Curriculum Vitae of

MAMAUAG, JEFFERSON G.

<Block 66 Lot 24 Nuvista Subdivision Barangay Santo Cristo City of San Jose Del Monte Bulacan>
mamauag.248577@sjdelmonte.sti.edu.ph
0931096819

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	**August 2023**	**STI College San Jose Del Monte**
Vocational/Technical		
High School	**May 2016-2023**	**Yverdon De Pestalozzi School**
Elementary	**June 2010-2016**	**Placido Del Mundo Elementary School**

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
July 2021-2023	Code Camp Society	Member
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	CS 2nd Year representative
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
MS Word	**Intermediate**	**September 2016**
Java Programming	**Intermediate**	**December 2023**
Python Programming	**Intermediate**	**April 2025**

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	**National Service Training Program**
February 2024	**GitGud**

Curriculum Vitae of

Tayag, Jose Mari Christian V.

Block 34, Lot 5, Saint Therese Phase, Deca Homes Subdivision, Barangay Loma de Gato, Marilao, Bulacan
tayag.334839@sjdelmonte.sti.edu.ph
+639166248648

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	**August 2023**	**STI College San Jose Del Monte**
Vocational/Technical		
High School	**May 2014-2020**	**Caloocan National Science and Technology Highschool**
Elementary	**June 2008-2014**	**ST. Anne Power Academy-Marilao, inc.**

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	Member
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
Ux/Ui Design	Intermediate	September 2020
Python Programming	Intermediate	March 2018
Rust Programming	Intermediate	December 2024

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	**National Service Training Program**
February 2024	**GitGud**
	CodeFest

Curriculum Vitae of

Accad, Glenn Marco B.

Block 32, Lot 17 Phase 6C Towerville, Barangay Gaya - Gaya, Towerville, San Jose Del Monte Bulacan
Accad.355761@sjdelmonte.sti.edu.ph
+639981847987

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2023	STI College San Jose Del Monte
Vocational/Technical		
High School	May 2017-2023	Graceville National High school
Elementary	June 2011-2017	Marangal Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience Job Title	Name and Address of Company or Organization
February 2023 – May 2023	Student Assistant	San Jose Academy of Bulacan

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	Member
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
MS Office	Intermediate	June 2011
Java Programming	Intermediate	October 2023
C# Programming	Intermediate	May 2024

TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	**National Service Training Program**
February 2024	**GitGud**
	CodeFest

Curriculum Vitae of

Gallano, Adrian V.

1067 Malaya Street, Sapang Alat Road, Barangay 181, Pangarap Village, Caloocan City, NCR
gallano.360632@sjdelmonte.sti.edu.ph
+63927208694

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2023	STI College San Jose Del Monte
Vocational/Technical		
High School	July 2017-2023	Immaculada Conception Colleges
Elementary	August 2011-2017	Our Lady of Mercy School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience Job Title	Name and Address of Company or Organization

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2023-2024	Intellitech	Member
August 2024-2025	Intellitech	Member
August 2025-2026	Intellitech	Member

SKILLS

SKILLS	Level of Competency	Date Acquired
MS Office	Intermediate	June 2018
Java Programming	Intermediate	October 2023
C# Programming	Intermediate	September 2024

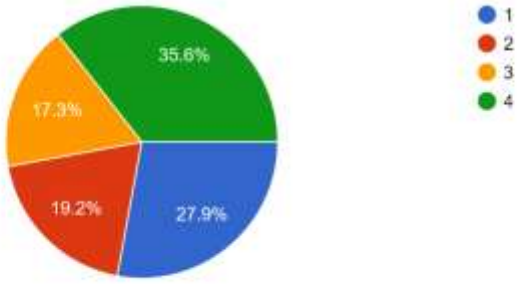
TRAININGS, SEMINARS OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar or Workshop
August 2023	National Service Training Program
February 2024	GitGud

Survey

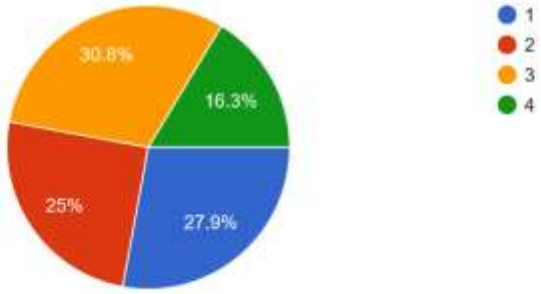
On a scale of 1–4, how familiar are you with the Lua programming language?

104 responses



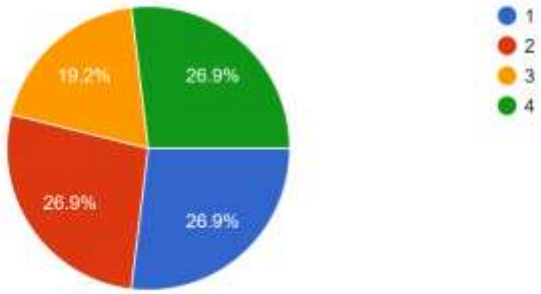
On a scale of 1–4, how would you rate the difficulty of understanding Lua's syntax and structure?

104 responses



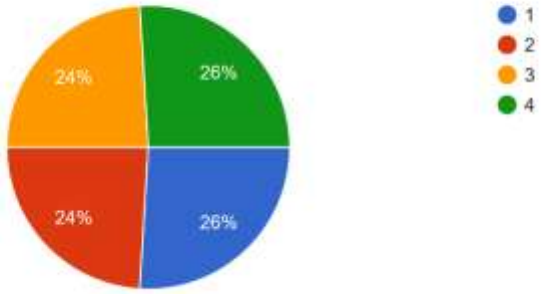
On a scale of 1–4, how would you rate your ability to learn Lua concepts through interactive or hands-on activities?

104 responses



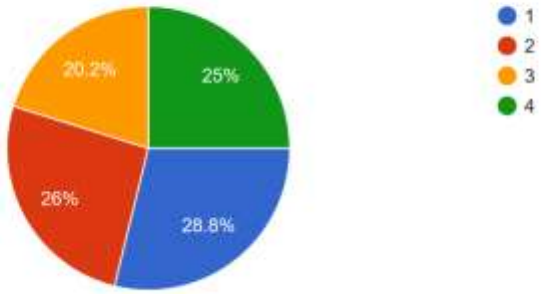
On a scale of 1–4, how helpful is practical, trial-and-error learning when understanding Lua?

104 responses



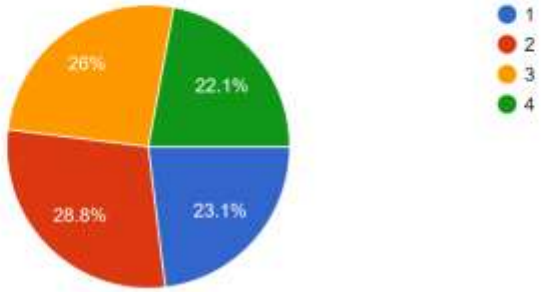
On a scale of 1–4, how helpful is visual feedback in understanding Lua programming concepts?

104 responses



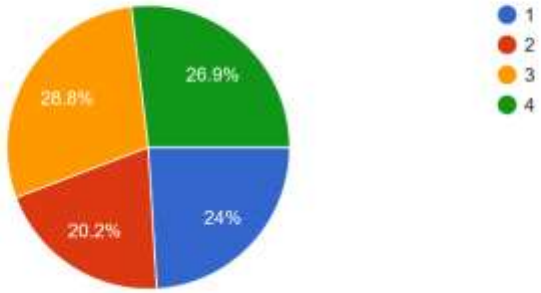
On a scale of 1–4, how clearly can you connect your Lua inputs (code) to the outputs or changes you see?

104 responses



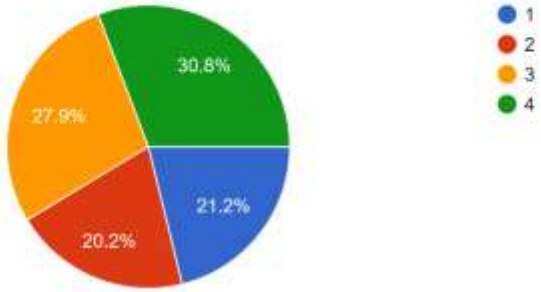
On a scale of 1–4, how helpful is immediate error feedback in understanding Lua concepts?

104 responses



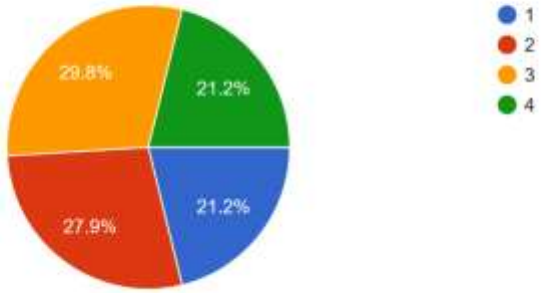
On a scale of 1–4, how interested are you in using supplementary tools (e.g., interactive platforms or games) to help you learn Lua more effectively?

104 responses



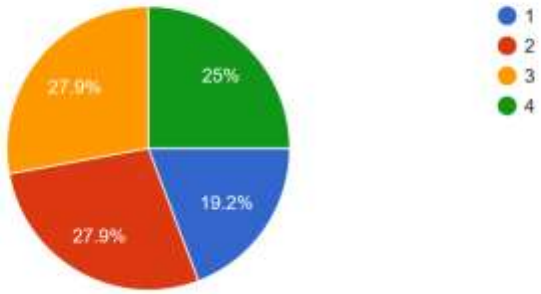
On a scale of 1–4, how familiar are you with the Python programming language?

104 responses



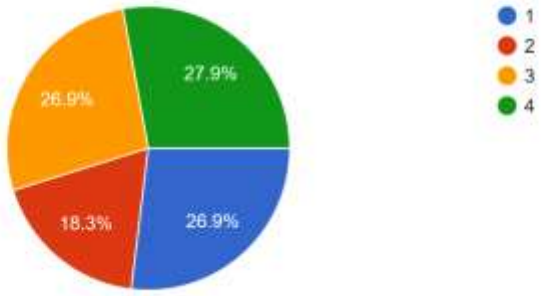
On a scale of 1–4, how would you rate the difficulty of understanding Python's syntax and structure?

104 responses



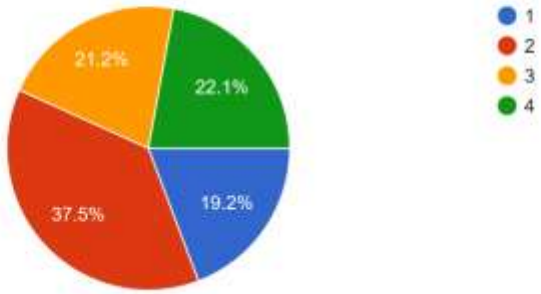
On a scale of 1–4, how would you rate your ability to learn Python through interactive or hands-on activities?

104 responses



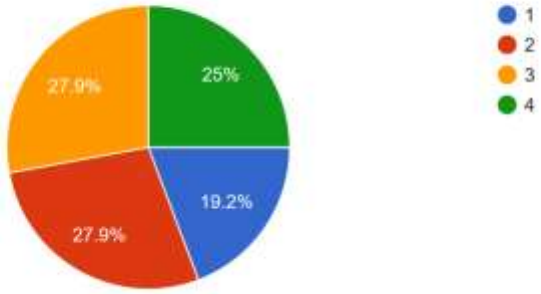
On a scale of 1–4, how helpful is practical, trial-and-error learning when understanding Python concepts?

104 responses



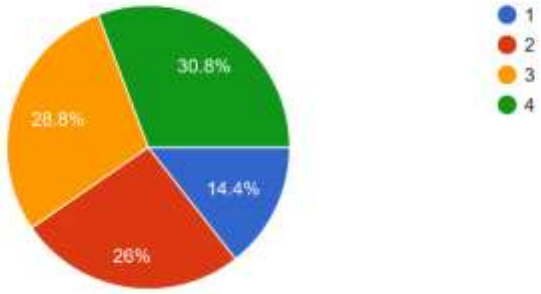
On a scale of 1–4, how helpful is visual feedback in understanding Python programming concepts?

104 responses



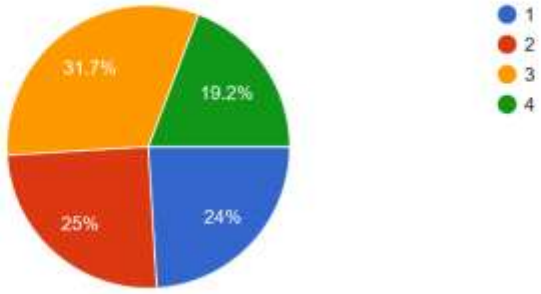
On a scale of 1–4, how clearly can you connect your Python code to the output or changes you see?

104 responses



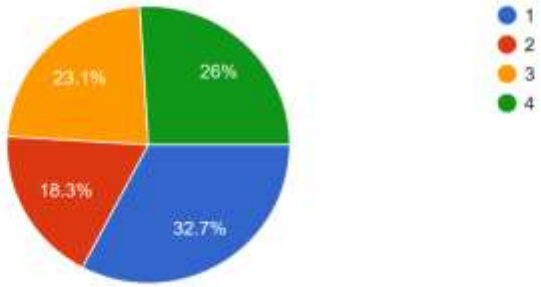
On a scale of 1–4, how helpful is immediate error feedback in understanding Python concepts?

104 responses



On a scale of 1–4, how interested are you in using supplementary tools (e.g., interactive platforms or games) to help you learn Python more effectively?

104 responses



ADVISER'S ACCEPTANCE FORM

NAME OF PROPONENTS:	**Glenn Marco B. Accad** **Adrian V. Gallano** **Jefferson G. Mamauag** **Jose Mari Christian V. Tayag**
APPROVED RESEARCH TITLE:	**SandEE: AN EDUCATIONAL MULTI-PLATFORM 2D OPEN WORLD SURVIVAL CRAFTING GAME FOR LUA AND PYTHON FOR ICT AND STEM STUDENTS OF YVERDON DE PESTALLOZI SCHOOL INC.**
AREA OF STUDY:	**Cross-Platform Application, Game-Based Learning, Programming**

CONFORME:

Norvin Shaq F. Lintag
Thesis Adviser

APPROVED BY:

Enoch James C. Tayanguna **Date: 13 September 2025**
Thesis Coordinator

NOTED BY:

Deffrey Josh A. Austria
Program Head

REFERENCES

Pohl, M., Rester, M., Judmaier, P. (2009). Interactive Game Based Learning: Advantages and Disadvantages. In: Stephanidis, C. (eds) Universal Access in Human-Computer Interaction. Applications and Services. UAHCI 2009. Lecture Notes in Computer Science, vol 5616. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-02713-0_10

D. Zhao, C. H. Muntean, A. E. Chis, G. Rozinaj and G. -M. Muntean, "Game-Based Learning: Enhancing Student Experience, Knowledge Gain, and Usability in Higher Education Programming Courses," in IEEE Transactions on Education, vol. 65, no. 4, pp. 502-513, Nov. 2022, doi: 10.1109/TE.2021.3136914.

Lampropoulos, G. (2023). Educational benefits of digital game-based learning: K-12 teachers' perspectives and attitudes. Advances in Mobile Learning Educational Research, 3(2), 805-817. https://doi.org/10.25082/AMLER.2023.02.008

Becker, B. A., Denny, P., Finnie-Ansley, J., Luxton-Reilly, A., Prather, J., & Santos, E. A. (2023). Programming is hard—or at least it used to be: Educational opportunities and challenges of AI code generation. In Proceedings of the 54th ACM Technical Symposium on Computer Science Education V. 1 (SIGCSE 2023) (pp. 500–506). Association for Computing Machinery. https://doi.org/10.1145/3545945.3569759

Brumbaugh, Z., & Leithner, M. (2023). The Lua programming language. In Programming languages: Concepts and implementations (pp. 45–60). Springer. https://doi.org/10.1007/979-8-8688-0060-3_3

Crudu, V., & MoldStud Research Team. (2024, September 27). Are Lua developers in high demand in the job market? MoldStud. https://moldstud.com/articles/p-are-lua-developers-in-high-demand-in-the-job-market

Ierusalimschy, R., Figueiredo, L. H. de, & Celes, W. (2025). The evolution of Lua, continued. Journal of Computer Languages, 83, 101326. https://doi.org/10.1016/j.cola.2025.101326

Lin, H.-C. K., Tseng, C.-H., & Chen, N.-S. (2025). Enhancing programming education: The impact of AI-based pedagogical agents on student self-efficacy, engagement, and learning outcomes. Educational Technology & Society, 28(2), 279–294. https://www.jstor.org/stable/48827987

Python, W. (2021). Python: Python releases for Windows. Python Software Foundation. https://www.python.org/downloads/windows/

Viduka, D., Kraguljac, V., & Ličina, B. (2021). A comparative analysis of the benefits of Python and Java for beginners. Quaestus, (19), 318–327.